

1.) Define Engineering Mechanics?

Engineering mechanics may be defined as the branch of science which deals with the behaviour of a body with the state of rest or motion, subjected to the action of forces.

2.) Define Rigid bodies?

The body which will not deform or the body in which deformation can be neglected in the analysis are called as rigid bodies.

3.) Define Deformable bodies?

If the internal stresses developed in a body are to be studied, the deformation of the body should be considered. This field of mechanics is called mechanics of deformable bodies.

4.) Distinguish between statics and dynamics?

Statics: The mechanics of the rigid bodies dealing with the bodies at rest is termed as statics.

Dynamics: The mechanics of the rigid bodies dealing with bodies in motion is called dynamics.

5.) Distinguish between kinetics and kinematics?

Kinetics deals with the forces causing motion. Kinematics is the study of motion of bodies without considering the forces causing motion.

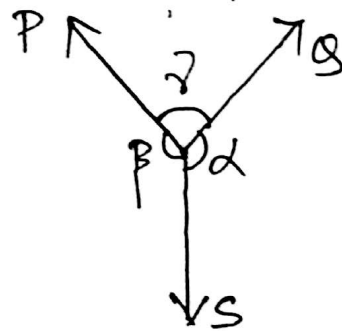
6.) State Principle of Transmissibility ?

According to this law, the state of rest or motion of the rigid body is unaltered if a force acting on the body is replaced by another force of the same magnitude and direction but acting anywhere on the body along the line of action of the replaced force.

7.) State Lami's theorem ?

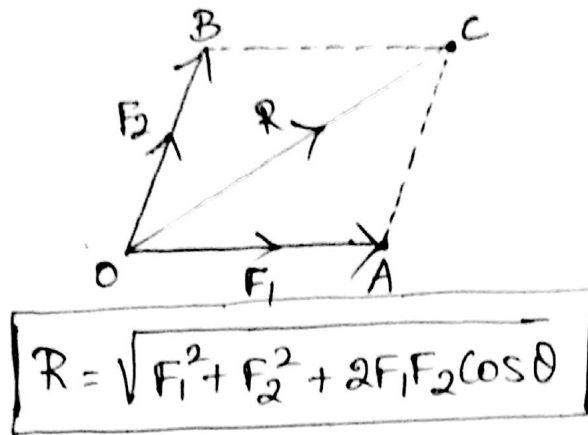
It states that "If three coplanar forces acting at a point be in equilibrium, then each force is proportional to the sine of the angle between the other two."

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{S}{\sin \gamma}$$



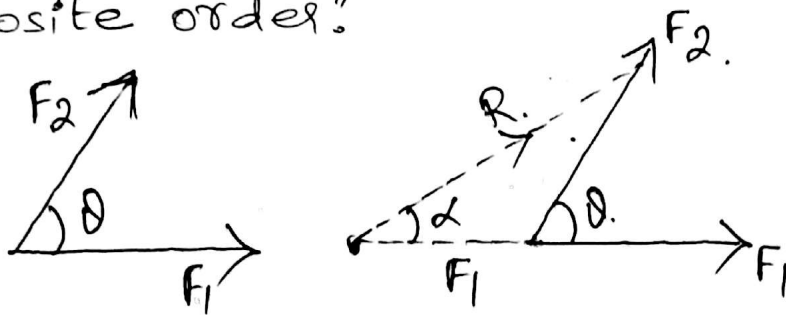
8.) State Parallelogram law of forces ?

It states that "If two forces acting simultaneously at a point be represented in magnitude and direction by the two adjacent sides of a parallelogram, then the resultant of these two forces is represented in magnitude and direction by the diagonal of that parallelogram which passes through the point of intersection of the two sides representing the forces."



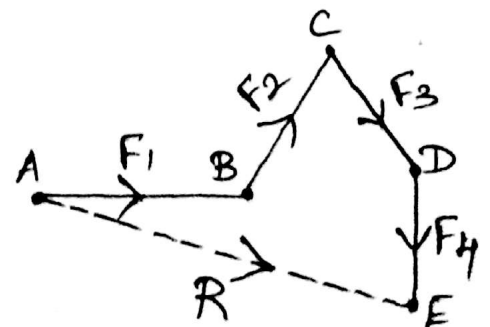
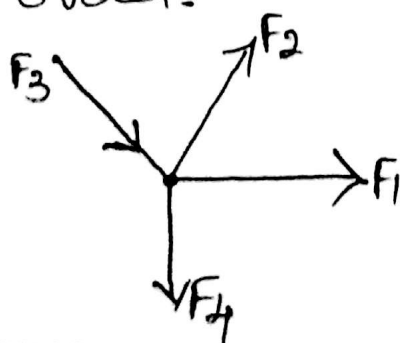
9.) State Triangle law of forces.

"If two forces (F_1, F_2) acting simultaneously on a particle can be represented by the two sides of a triangle (in magnitude and direction) taken in order, then the third side (closing side) represents the resultant in the opposite order."



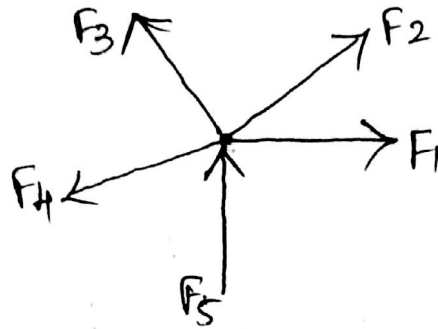
10.) State Polygon law of forces.

"If a number of concurrent forces acting simultaneously on a particle are represented in magnitude and direction by the sides of Polygon taken in order, then the resultant of this system of forces is represented by the closing side of the Polygon in the opposite order."

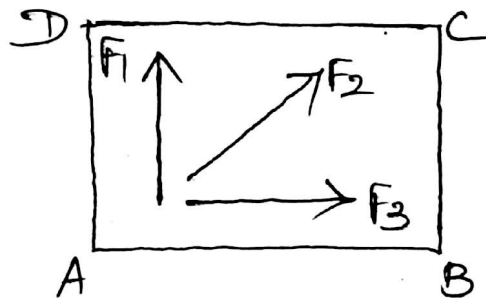


11.) Differentiate between Concurrent force and Coplanar force.

Concurrent force: Line of action of all forces pass through a single point.

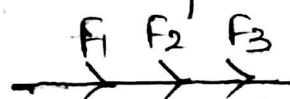


Coplanar force: If all the forces in a system lie in a single plane, it is called Coplanar force system.



12.) Differentiate between Collinear and Parallel force.

Collinear force: The forces which act on a common line of action are called collinear force.



like collinear.



unlike collinear

Parallel force: Line of action of forces are parallel to each other.



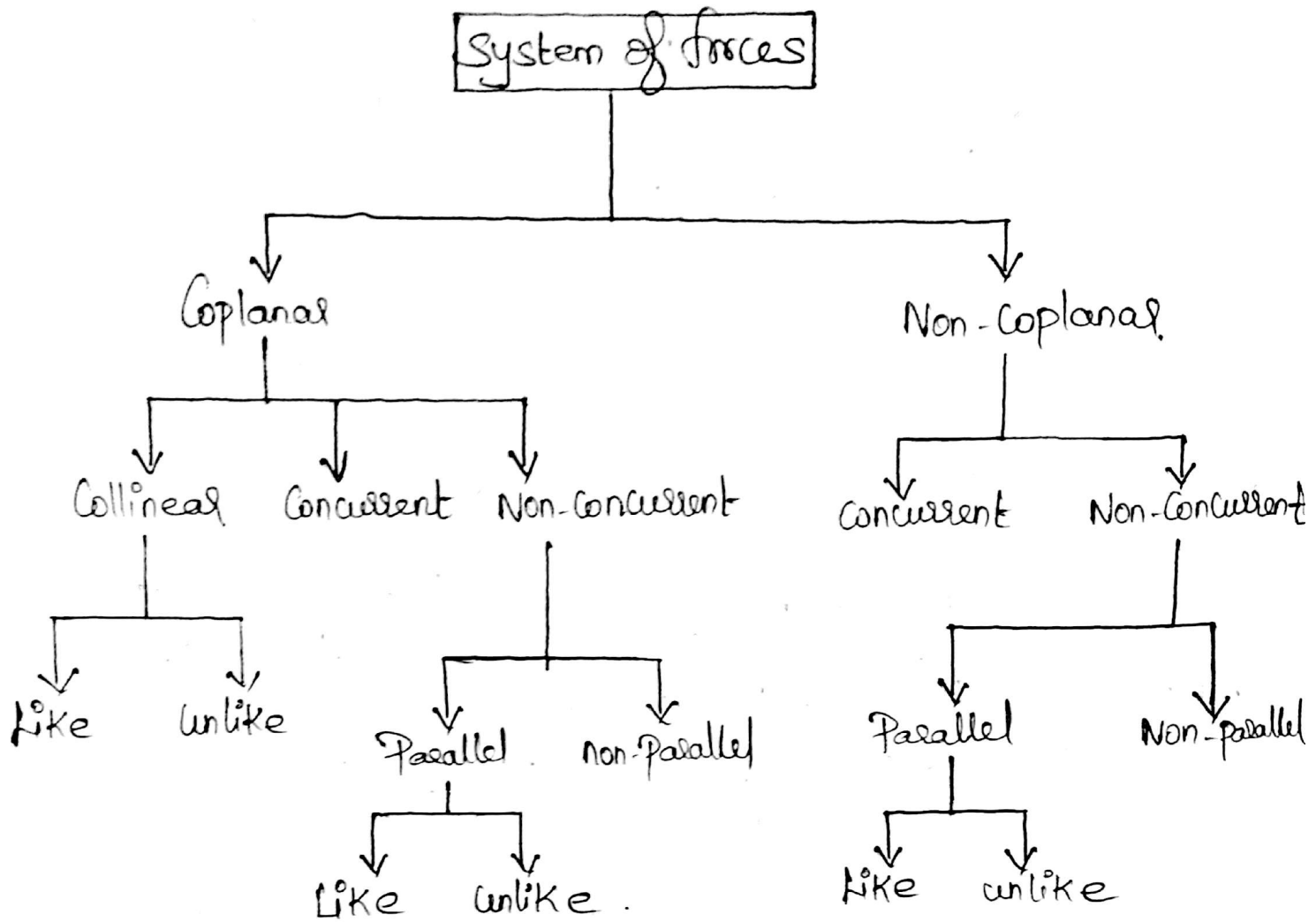
like parallel force



unlike parallel force.

13.) Classify various types of forces.

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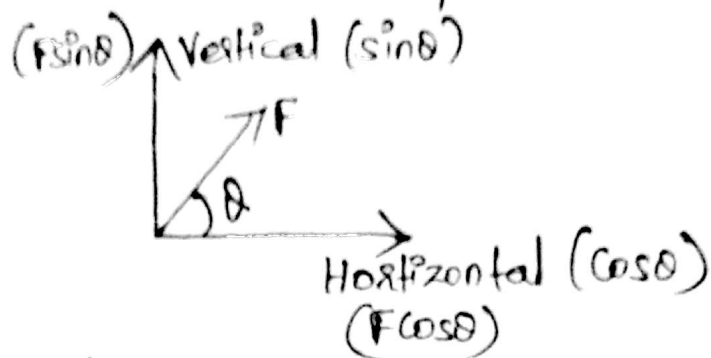
14.) Define Resultant force.

If a number of forces acting on a Particle simultaneously are replaced by a single force, which could produce the same effect as produced by the given forces that single force is called Resultant force.

$$\begin{array}{c} P \uparrow \\ Q \uparrow \\ S \uparrow \end{array} = \uparrow R = (P + Q + S)$$

15.) What is Resolution of a force.

Splitting up a force into components along the fixed reference axes is called Resolution of a force.



16.) Differentiate Equilibrium and Equilibrant?

Equilibrium: A body is said to be in equilibrium when the resultant of the force system acting on it is zero.

Equilibrant: The force which brings the particle to equilibrium is called Equilibrant.

17.) Write the necessary and sufficient conditions of equilibrium?

For Equilibrium condition of force system, the resultant is zero

$$(i.e) R = 0$$

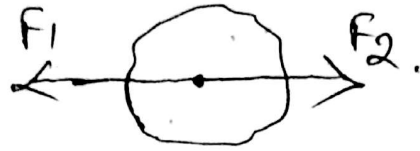
$$\text{But } R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

Hence, both ΣH and ΣV are to be zero for equilibrium condition.

18.) What is two force Equilibrium principle?

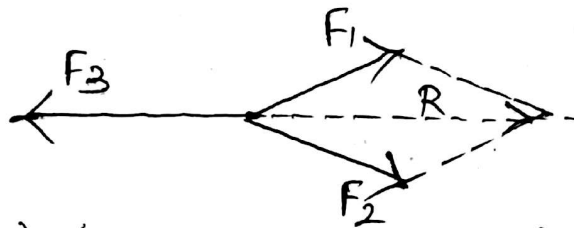
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"If a body is subjected to two forces, then the body will be in equilibrium if the two forces are collinear, equal and opposite."



19.) What is Three force Equilibrium principle?

"If a body is subjected to three forces, then the body will be in equilibrium, if the resultant of any two forces is equal, opposite and collinear with the third force."



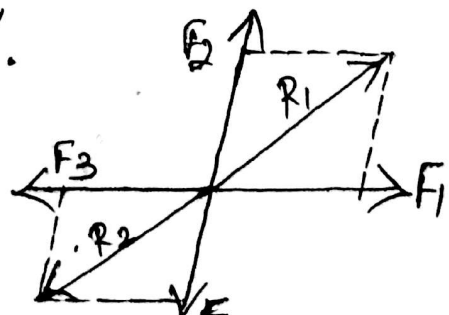
R is the resultant of F_1 and F_2 and also

$$R = F_3.$$

20.) What is Four force Equilibrium principle?

"If a body is subjected to four forces, then the body will be in equilibrium, if the resultant of any two forces is equal, opposite and collinear with the resultant of the other two."

$$R_1 = R_2$$



Q1.) Distinguish between Freebody and Freebody diagram?

Freebody: It is the body isolated from all other members which are connected to this body.

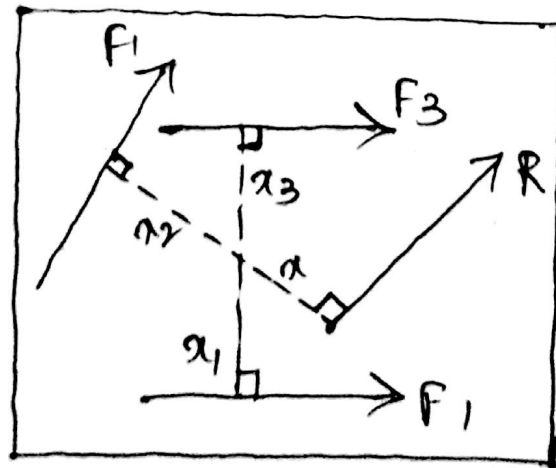
Freebody diagram: The sketch showing all the forces (Both external forces and reactions) and moments acting on the body is called as Freebody diagram.

Q2.) State Varignon's theorem.

Varignon's theorem is also known as theorem of moments. It states that "algebraic sum of the moments of any number of forces about any point in their plane is equal to the moment of their resultant about the same point."

$$\sum M = R \alpha$$

$$\sum M = F_1 x_1 + F_2 x_2 + F_3 x_3 + \dots + F_n x_n$$



Varignon's theorem.

23.) Define Moment of a force about a point. [Define Moment].

Moment of a force about a point is defined as the product of the force and perpendicular distance of the line of action of the force from that point.

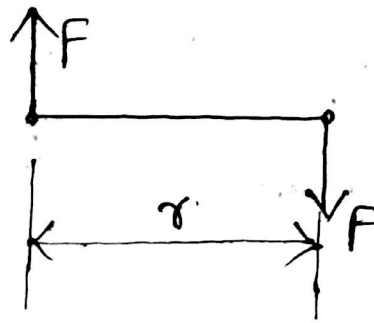


Moment of the force (F) about 'O' is given by

$$M_O = F \times r \quad [\text{i.e. Force} \times \text{Perpendicular distance}]$$

24.) Define Couple & how it is classified.

Two forces F and $-F$ having the same magnitude, parallel lines of action and opposite sense are said to form a couple.



Couple is classified into two types based on its nature

1. Clockwise Couple
2. Anticlockwise Couple.

Q5.) Define force and what are the characteristics of a force.

⇒ Force is nothing but a push or pull.

⇒ Force is an agent which changes or tends to change the state of rest or of uniform motion of a body upon which it acts.

Characteristics of a force:

- (i) Magnitude
- (ii) Line of action
- (iii) Direction.

Q6.) What are the applications of Engineering Mechanics.

(i) Working Principles of Engines.

(ii) Study of missiles and aeroplanes.

(iii) External forces acting on a ship (or) boat in water.

NOTE: See problems in Resultant force of two concurrent forces. [2 marks]

Formula:

$$\Rightarrow R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

$$\Rightarrow \tan \alpha = \frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta}$$